

Support for LEO-PNT Constellation Data in IGS Data and Product Files

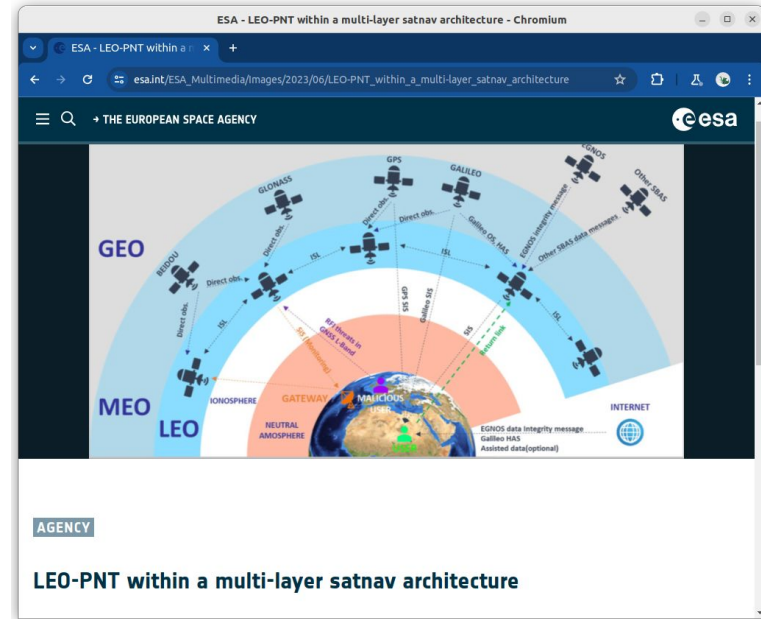
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IGS Workshop 2024
July 1st 2024, Session 1

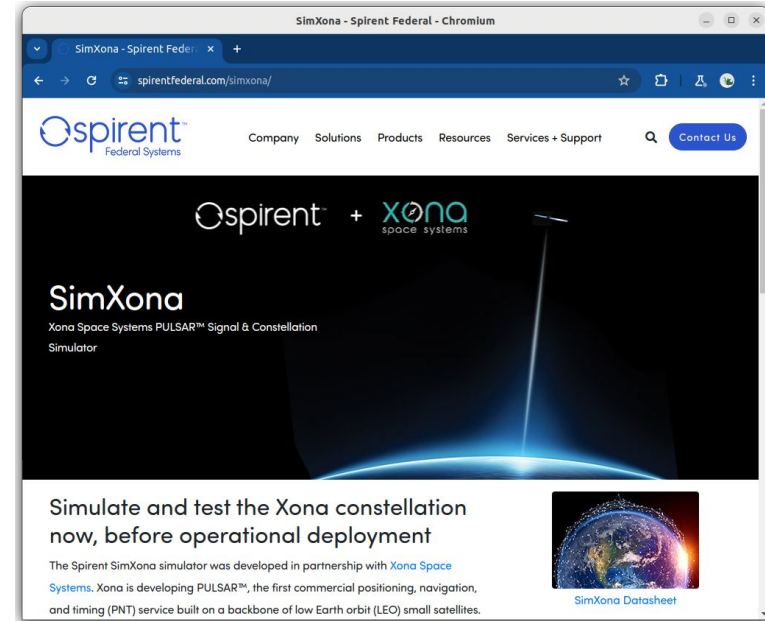
Motivation

- Multi-layer (e.g. LEO) PNT is gaining momentum in the GNSS world
- Institutional: ESA LEO-PNT In orbit demonstration mission, NSSTC/UAE
- Industry (commercial PNT): Xona, Centispace, Trustpoint, Spirent, ...
- Industry is already moving forward to advance technology
 - Positioning Engines
 - Research applications
- Need for a standardized format to exchange data!



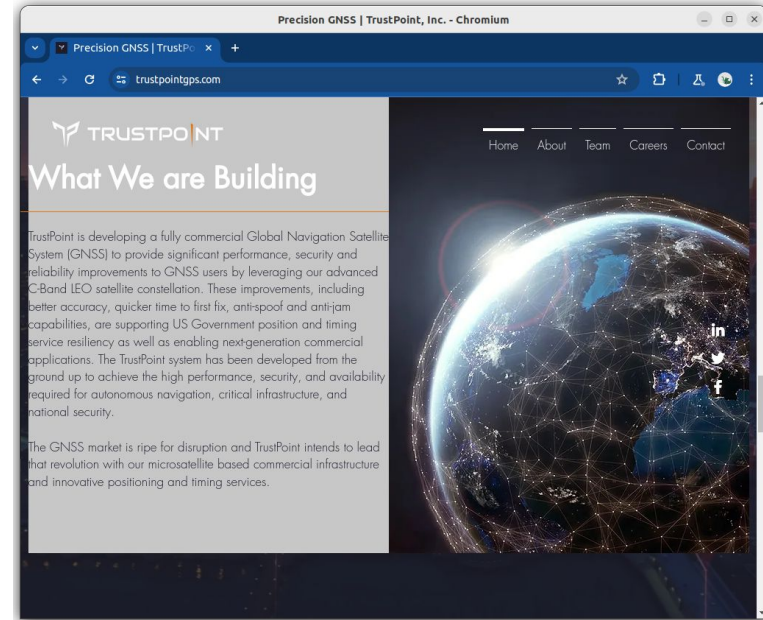
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RINEX Navigation

- “Easiest to adapt” in RINEX version 4
- Needs agreement on constellations to include as well as fields for each constellation (e.g. signal biases)

Rinex 4 “EPH” allows arbitrarily long Sat ID descriptor (no need to place it in the data block)

LEO constellation type	Constellation	LEO satellite id
L	Generic LEO-PNT constellation	custom
P	Xona	NORAD id
Y	Geely	NORAD id
X	Starlink	NORAD id
O	Oneweb	NORAD id
V	Spire	NORAD id
Z	Centispace	NORAD id

NORAD id (5 digit) could be used. Mega-constellations may reach thousand satellites (e.g. Starlink)

Diagram illustrating the RINEX Navigation data block structure with annotations:

```

> EPH z02434
2023 11 02 11 10 24 4.887547846484e-05 4.303787327448e-12 0.000000000000e+00
0.000000000000e+00 0.000000000000e+00 1.989675347274e-09 4.642884733583e+00
0.000000000000e+00 1.811000000000e-04 0.000000000000e+00 2.835776281159e+03
3.858240000000e+05 0.000000000000e+00 5.882354736496e+00 0.000000000000e+00
9.288031413876e-01 0.000000000000e+00 1.642393223370e+00 0.000000000000e+00
0.000000000000e+00 0.000000000000e+00 2.286000000000e+03 0.000000000000e+00
0.000000000000e+00 0.000000000000e+00 1.027763024283e-09 0.000000000000e+00
-9.724269021370e-10 0.000000000000e+00 0.000000000000e+00 0.000000000000e+00
  
```

Annotations:

- Time of ephemeris:** Points to the date and time fields (2023 11 02 11 10 24).
- Satellite clock:** Points to the clock bias and rate fields (4.887547846484e-05, 4.303787327448e-12).
- Orbital parameters:** Points to the semi-major axis, eccentricity, inclination, and other orbital elements.
- Signal biases:** Points to the signal bias fields (2.286000000000e+03, 1.027763024283e-09).

RINEX Observation

- Critical **limitation**: 2-digit for Satellite ID
- Needs to maintain **consistency** with proposed extension for RINEX NAV (5-digit satellite ID)
- After several iterations, opted to add a “breaking change” where the parser will have to check first letter to select the appropriate parser (e.g. decide how to parse satellite ID, necessary fields, ...).

OBSERVATION DATA FILE - DATA RECORD DESCRIPTION (LEO)	
DESCRIPTION	FORMAT
EPOCH record	
- Record start identifier : ">"	A1
- Epoch:	
- year (4 digits)	1X,I4
- month, day, hour, min (2 digits)	4(1X,I2.2)
- sec	F11.7
- Epoch flag [*]	2X,I1
- Number of satellites in current epoch	I3
- (reserved)	6X
- receiver clock offset (seconds, optional)	F15.12
[*] EPOCH FLAG follow the same definition as RINEX 4.0	
OBSERVATION record for LEO-PNT satellites [**]	
- Satellite number (LEO system dependent, e.g. NORAD ID)	A1,I5
- m fields of observation data in the same sequence as given in the "SYS / # / OBS TYPES" header record	m(F14.3,
- LLI each obs.type (same seq	I1,
- Signal strength as given in header)	I1)
[**] Observation records for MEO GNSS are the same as RINEX 4	

RINEX Observation

- Example

```

4.99      OBSERVATION DATA      M (MIXED)      RINEX VERSION / TYPE
rdd 76.46.1-r1      Rokubun      20240613 105750 UTC      PGM / RUN BY / DATE
SIM                                          MARKER NAME
                                          MARKER NUMBER
                                          MARKER TYPE
                                          OBSERVER / AGENCY
                                          REC # / TYPE / VERS
                                          ANT # / TYPE
                                          APPROX POSITION XYZ
                                          ANTENNA: DELTA H/E/N
                                          TIME OF FIRST OBS
                                          TIME OF LAST OBS
                                          SYS / # / OBS TYPES
                                          SYS / # / OBS TYPES
                                          SYS / # / OBS TYPES
                                          END OF HEADER

                                argos                                76.46.1-r1

                                0.0000                                0.0000                                0.0000
                                0.0000                                0.0000                                0.0000
2023      11      01      19      52      17.00000000      GPS
2023      11      01      20      13      57.00000000      GPS
G      4 C1C L1C D1C S1C
E      4 C1C L1C D1C S1C
Z      4 C1C L1C D1C S1C

> 2023 11 01 19 52 17.00000000 0 15
E13 25878172.51301 135990714.41801 -2235.82001
E21 24723841.81401 129924665.65301 -2178.18601
E26 23323688.40101 122566809.80701 -375.24701
E31 26757149.28001 140609768.50701 2545.19701
E33 25223059.64801 132548073.07601 1626.59801
G05 22952763.20301 120617584.73401 3596.63401
G06 22062030.79501 115936754.33601 -2652.35101
G11 20540902.96801 107943173.65301 -851.69301
G12 20392939.80101 107165621.96401 475.05801
G19 23163152.88101 121723189.95101 -2373.50601
G20 20884593.55701 109749279.88701 1663.99601
G24 24918715.92301 130948736.00901 -2767.12401
G25 22080762.26401 116035188.87301 2581.21401
G29 25095906.90201 131879880.88601 2656.54601
Z47817 3913093.12601 20563443.16801 27481.19301

```

SP3 format

- Problem statement: Similar than RINEX, 2-digit satellite ID
- Required modifications will have larger impact:
 - Satellite block in header
 - Satellite ID in PV rows
- Example: LEO orbit for Spire's LEMUR satellites.
 - ID specified in the file name:
leoOrb_2021.004.125.16_0001.0001_sp3
 - Uses **L99** inside the file (single satellite). Only 8 satellites would fit in each row of the satellite header block, instead of 17.

```
#cV2021 1 4 8 16 53.00000000 478 IGS08 FIT UCAR
## 2139 116213.00000000 60.00000000 59218 0.3450578703705
+ 1 L99 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
+ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
+ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
+ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
++ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
++ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
++ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
++ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
++ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
%c cc GPS ccc cccc cccc cccc cccc ccccc ccccc ccccc ccccc
%c cc cc ccc ccc cccc cccc cccc cccc ccccc ccccc ccccc ccccc
%f 00.00000000 00.0000000000 00.000000000000 00.0000000000000000
%i 0000 0000 0000 0000 000000 000000 000000 000000 0000000000
/*
/*
/*
/*
* 2021 1 4 8 16 53.00000000
PL99 5524.274281 3163.274083 -2698.438729 999999.999999
VL99 -17743.894636 -27918.462889 -69221.257287 999999.999999
```

Open questions

- Should we go for a RINEX format specific to LEO systems only (rather than mixing it with currently existing GNSS MEO systems?)
- Reference signal for satellite clock in RINEX Nav? (dual freq. Iono-combination? Which signals?)
- Description of orbital parameters in LEO orbit (larger perturbations): Increase update rate (allowed by the format), increase grade of sine/cosine coefficients (may require update of the format, ... Alternatively, should we use GLONASS- /SBAS- like ephemeris (XYZ)
- How to assign letters for satellite constellation. Any limitations?
- Signal biases in RINEX Nav: To be defined on a per-constellation basis. Number of frequencies and channels (data? pilot?...). Example: Xona frequencies are public, but not its signal structure.

Conclusions

Extension of RINEX and SP3 format proposed

- RINEX format is widely used not only in academia, but also in industry
- Development of Multi-layer PNT positioning engines (and other processing tools) are already underway
- Satellites for [LEO-PNT already in place](#)
- [ESA's LEO-PNT missions in-orbit before 2027](#)
- Therefore, standardization is important to foster development and knowledge exchange.

Ongoing and next steps:

- Gather feedback from relevant stakeholders (academia, receiver manufacturers, simulation, ...) to address open questions.
- Github repo with format proposal: https://github.com/rokubun/rinex_leo (access upon request)



Thank you!

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